

2024.

$$Q1. (a) \frac{C(s)}{R(s)} = \frac{K(s)G(s)}{1+K(s)G(s)} = \frac{\frac{s}{(s+5)(s+2)}}{1+\frac{s}{(s+5)(s+2)}} = \frac{s}{s^2+8s+10}$$

Poles from  $s^2+8s+10=0$

$$\Rightarrow P_{1,2} = -4 \pm \sqrt{6} \Rightarrow P_1 = -1.55 \quad P_2 = -6.45$$

negative and real

Two poles on the real axis are both negative

so it's stable, not oscillatory, overdamped.

exponentially decreasing

$$(b). G(s) = \frac{10(s+2.5)(s+5.6)}{(s+4)(s+7)} = \frac{C(s)}{R(s)}$$

For unit step:  $R(s) = \frac{1}{s}$

$$\Rightarrow C(s) = \frac{10(s+2.5)(s+5.6)}{s(s+4)(s+7)} = \frac{A}{s} + \frac{B}{s+4} + \frac{C}{s+7}$$

$$= \frac{A(s+4)(s+7) + Bs(s+7) + Cs(s+4)}{s(s+4)(s+7)}$$

$$s = -2.5 \Rightarrow A \times 1.5 \times 4.5 + B \times (-2.5) \times 4.5 + C \times (-2.5) \times 1.5 = 0$$

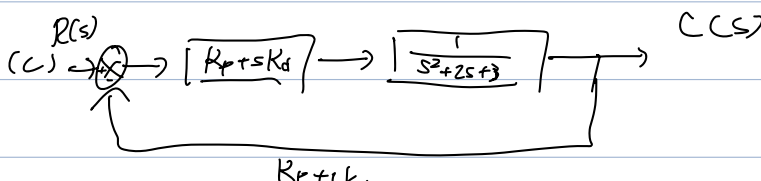
$$s = -5.6 \Rightarrow A \times (-1.6) \times 1.4 + B \times (-5.6) \times 1.4 + C \times (5.6) \times (-1.6) = 0$$

$$s = 0 \Rightarrow A \times 4 \times 7 = 10 \times 2.5 \times 5.6$$

$$\Rightarrow \begin{cases} A = 5 \\ B = 2 \\ C = 3 \end{cases}$$

$$\Rightarrow C(s) = \frac{5}{s} + \frac{2}{s+4} + \frac{3}{s+7}$$

$$\Rightarrow c(t) = 5 + 2e^{-4t} + 3e^{-7t} \quad \text{for a unit step input}$$



$$\frac{C(s)}{R(s)} = \frac{\frac{K_p + sK_d}{s^2 + 2s + 3}}{1 + \frac{K_p + sK_d}{s^2 + 2s + 3}} = \frac{K_p + sK_d}{s^2 + 2s + 3 + K_p + sK_d}$$

$$= \frac{K_d s + K_p}{s^2 + (2 + K_d)s + 3 + K_p}$$

characteristic eqn. compared with ...

gives:  $\begin{cases} 2 + K_d = 2\zeta\omega_n \\ 3 + K_p = \omega_n^2 \end{cases} \Rightarrow \begin{cases} \omega_n = \sqrt{3 + K_p} \\ \zeta = \frac{2 + K_d}{2\sqrt{3 + K_p}} \end{cases}$

settling time  $t_s = \frac{3}{\zeta\omega_n} = \frac{6}{2 + K_d} = 1.5 \text{ s}$

close-loop response  $C(s) = \lim_{s \rightarrow 0} s \cdot \frac{C(s)}{R(s)} = \frac{K_p}{3 + K_p} = 0.96$

$\Rightarrow \begin{cases} K_p = 72 \\ K_d = 2 \end{cases} \Rightarrow \begin{cases} \omega_n = 5\sqrt{3} \approx 8.66 \text{ rad/s} \\ \zeta = \frac{2\sqrt{3}}{15} \approx 0.23 \end{cases}$

(d) Routh criterion.

$$\begin{array}{c|ccc} s^4 & 1 & 2 & K \\ s^3 & 12 & 5 & 0 \\ s^2 & \frac{19}{12} & K & 0 \\ s^1 & c_1 & 0 & \\ s^0 & K & & \end{array}$$

$$b_1 = \frac{-1}{12} \begin{vmatrix} 1 & 2 \\ 12 & 5 \end{vmatrix} = -\frac{1}{12} (5 - 24) = \frac{19}{12}$$

$$b_2 = \frac{-1}{12} \begin{vmatrix} 1 & K \\ 12 & 0 \end{vmatrix} = -\frac{1}{12} (0 - 12K) = K$$

$$c_1 = \frac{-1}{19/12} \begin{vmatrix} 12 & 5 \\ 19/12 & K \end{vmatrix} = -\frac{12}{19} \left( 12K - \frac{19}{12} \times 5 \right)$$

$$\geq 0$$

$$\Rightarrow 12K - \frac{19}{12} \times 5 < 0$$

$$\Rightarrow K < \frac{19}{144} \approx 0.132$$

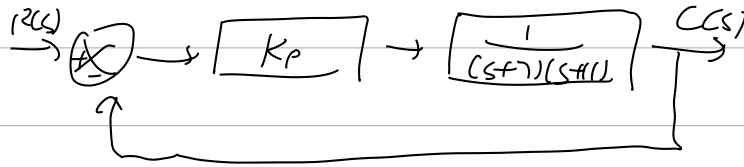
$$K > 0$$

$$\sim 0 < K < 0.132$$

$$c_2 = \frac{-1}{19/12} \begin{vmatrix} 12 & 0 \\ 19/12 & 0 \end{vmatrix} = 0$$

$$d_1 = -\frac{1}{c_1} \begin{vmatrix} \frac{19}{12} & K \\ c_1 & 0 \end{vmatrix} = -\frac{1}{c_1} (-c_1 K) = K$$

Q2. (a). If  $P(s) = 0$

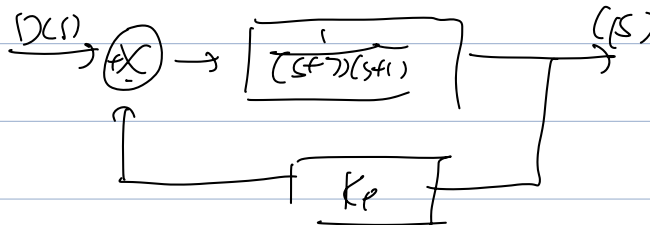


$$\frac{C(s)}{R(s)} = \frac{\frac{K_p}{(s+7)(s+11)}}{1 + \frac{K_p}{(s+7)(s+11)}} = \frac{K_p}{s^2 + 18s + 77 + K_p}$$

From ...  $P(s) = \dots$

$$C_{ss} = \lim_{s \rightarrow 0} s \cdot C(s) \cdot \frac{1}{s} = \frac{K_p}{77 + K_p}$$

(b) If  $P(s) = 0$



$$\frac{C(s)}{R(s)} = \frac{\frac{1}{(s+7)(s+11)}}{1 + \frac{K_p}{(s+7)(s+11)}} = \frac{1}{s^2 + 18s + 77 + K_p}$$

$$C_{ss} = \lim_{s \rightarrow 0} s \cdot C(s) \cdot \frac{1}{s} = \frac{1}{77 + K_p}$$

poles are real

(c).  $\Delta = 18^2 - 4(77 + K_p) \geq 0 \Rightarrow K_p \leq 4$

From (a), (b)

$$C_{ss} = \frac{4}{77 + 4} = \frac{4}{81} = 0.0494$$

$$C_{ss} = \frac{1}{81} = 0.0123$$

(d).  $C(s) = \frac{1}{s^2 + 18s + 80} = \frac{1}{(s+8)(s+10)} = \frac{A}{s+8} + \frac{B}{s+10}$

$$\Rightarrow C(xt) = \frac{1}{2} (e^{-8t} - e^{-12t}) = \frac{1}{2} \left( \frac{1}{s+8} - \frac{1}{s+12} \right)$$

Q3. (a) Let  $z=1$

$$G(s) = C(s) \cdot P(s)$$

$$= \frac{K(s+1)}{s(s+5)(s+2)}$$

There are 3 open loop poles: 0, -2, -5

1 open loop zero: -1

(ii). There are  $3-1=2$  asymptotes

$$\text{meeting up } \sigma = \frac{0-2-5-(-1)}{3-1} = -3$$

For the double poles:

$$\frac{dG(s)}{ds} = \frac{s(s+5)(s+2) - (s+1)(3s^2+14s+10)}{[s(s+5)(s+2)]^2} = 0$$

$$s^3 + 7s^2 + 10s$$

$$-2s^3 + (7-14-3)s^2 + (10-10+4)s - 10 = 0$$

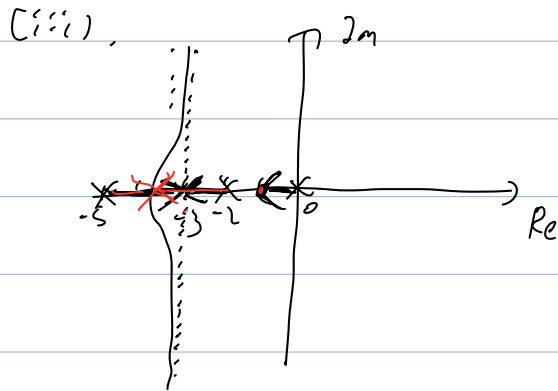
$$\Rightarrow -2s^3 - 10s^2 - 14s - 10 = 0$$

$$\Rightarrow s^3 + 5s^2 + 7s + 5 = 0$$

$$\Rightarrow s_1 = -3.359 \quad s_2 = -0.820 + 0.903j$$

$$s_3 = -0.820 - 0.903j$$

Therefore  $s = -3.359$ .



(iv). Stable for all  $K > 0$

When this system is overdamped.  $s = -3.36$

From  $\frac{K(s+1)}{s(s+5)(s+2)} + 1 = 0 \Rightarrow 0.315K + 1 = 0$

$\Rightarrow K = 3.175$

(b).  $X = \frac{0.2s - (-z_1)}{3 - 1} = \frac{z_1 - 7}{2} \Rightarrow z_1 = 7$

$G(s) = \frac{s + z_1}{s(s+5)(s+2)}$

$\frac{\partial G(s)}{\partial s} = \frac{s(s+5)(s+2) - (s+7)(3s^2 + 14s + 10)}{[s(s+5)(s+2)]^2} = 0$

$\Rightarrow s = -0.959$



$$c) z_1 < 0$$

$$1 + \frac{K(s+1)}{s(s+5)(s+2)}$$

$$\Rightarrow s^3 + 7s^2 + 10s + Ks + Kz_1 = 0$$

$$\Rightarrow s^3 + 7s^2 + (10+K)s + Kz_1 = 0$$

Routh-Hurwitz stability criterion

|       |      |      |   |                                                                                                                                                                                                                       |
|-------|------|------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $s^3$ | 1    | 10+K | 0 | $b_1 = \frac{1}{7} \begin{vmatrix} 1 & 10+K \\ 7 & Kz_1 \end{vmatrix}$<br>$= -\frac{1}{7} (Kz_1 - 70 - 7K)$<br>$c_1 = -\frac{1}{b_1} \begin{vmatrix} 7 & Kz_1 \\ b_1 & 0 \end{vmatrix}$<br>$= -\frac{1}{b_1} (-Kz_1)$ |
| $s^2$ | 7    | Kz_1 | 0 |                                                                                                                                                                                                                       |
| $s^1$ | b_1  | 0    |   |                                                                                                                                                                                                                       |
| $s^0$ | Kz_1 |      |   |                                                                                                                                                                                                                       |

$$b_1 < 0$$

$$Kz_1 - 70 - 7K > 0 \text{ for } \forall K > 0 \Rightarrow Kz_1$$

$$z_1 > \frac{70+7K}{K}$$

$$\frac{70}{K} + 7$$

NO!

Three methods that can be used to determine the value of K

• Routh-Hurwitz stability criterion

• Root locus analysis

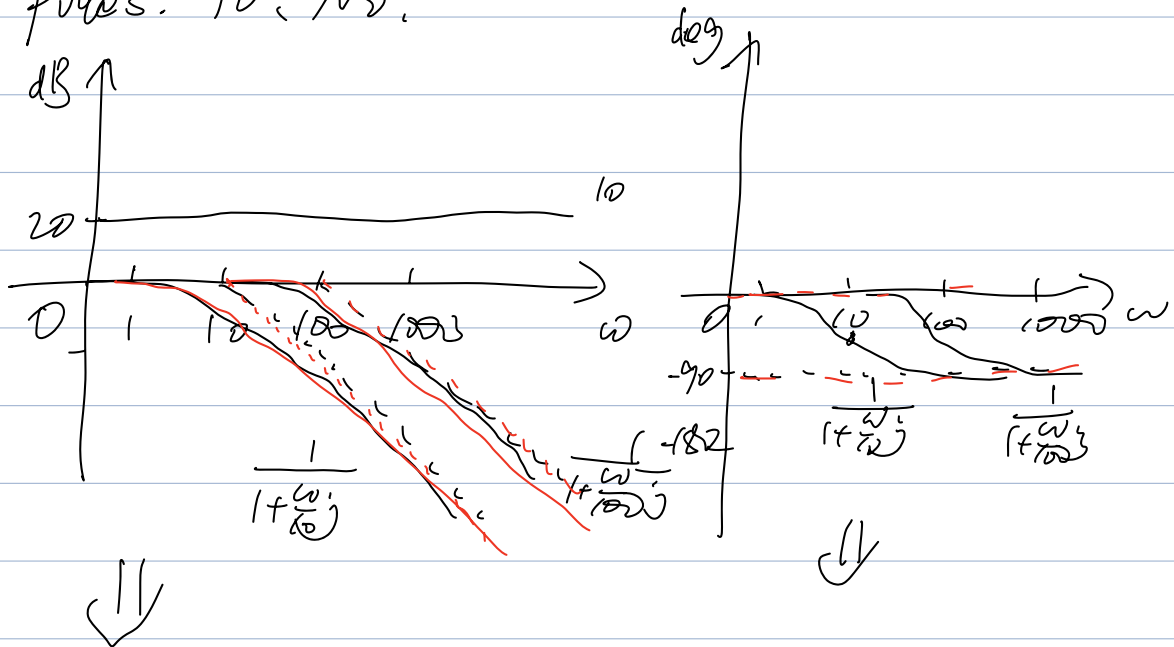
• Nyquist stability criterion (gain margin)

Q4. ch.  $C(s)P(s) = \frac{10}{(1+\frac{s}{10})(1+\frac{s}{100})}$

$\Rightarrow C(\omega)P(\omega) = \frac{10}{(1+\frac{\omega}{10}j)(1+\frac{\omega}{100}j)}$

Zeros: none.

Poles:  $10, 100$ .



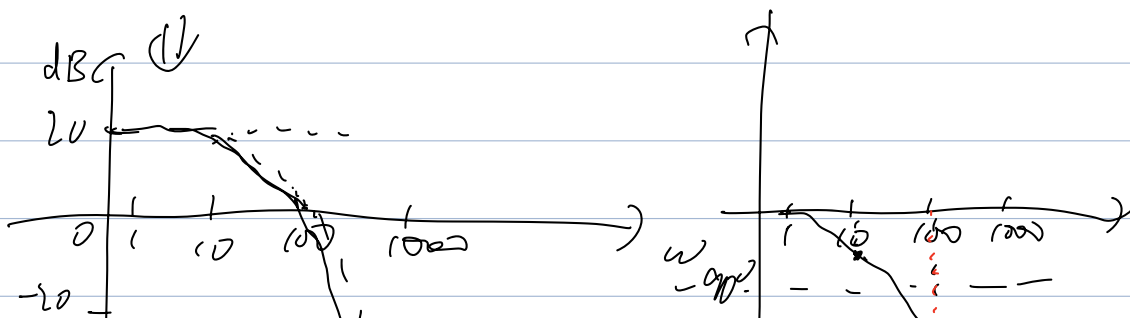
$0 < \omega < 10$     0 dB/dec

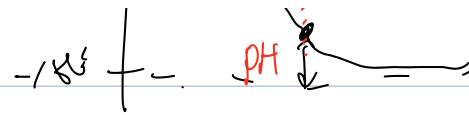
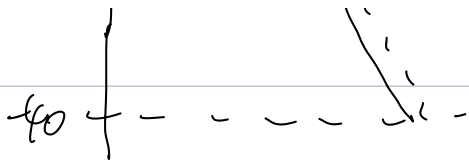
$\omega \approx 10$     -3 dB compensation

$10 < \omega < 100$     -20 dB/dec

$\omega \approx 100$     -3 dB compensation

$\omega > 100$     -40 dB/dec





$$45^\circ + 110^\circ = 155^\circ$$

$$180^\circ - (45^\circ + 90^\circ - 12^\circ)$$

$$124^\circ$$

$$= 56^\circ \approx 60^\circ$$

crit. gain margin =  $\infty$  infinite

phase margin  $\approx 56^\circ$